

Security for AI/ML models using Quantum Cryptography

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Related Discussion

Every organisation wants to protect its valuable digital assets like data and information, but what about AI algorithms? Successful exploitation of AI algorithms can cause financial loss, reputational damage, loss of competitive advantage, and loss of intellectual property.

For example, a malicious breach of pricing algorithm for an e-commerce company can result in attackers causing financial loss to the retailers/sellers and negatively impacting the platform's brand reputation.

The majority of organizations are struggling to address the challenge of securing models.

Problem Statement

Developing a simulation of securely transmitting AI model parameters (e.g., neural network weights) using Quantum Key Distribution (QKD) for key generation and post-quantum encryption for transmission.

QKD - technique used to securely share encryption keys between two parties by using the **principles of quantum mechanics**.

How it works (basic idea of BB84 protocol):

1. Quantum particles (**photons**) are sent from Alice to Bob.
2. Each photon is encoded in a **quantum state** (for example, different polarisations like $\leftrightarrow$ or $\nearrow$).
3. If Eve tries to measure these photons, the quantum states change due to the Heisenberg uncertainty principle.
4. Alice and Bob **compare** parts of their data over a public channel to detect any disturbance.
5. If no eavesdropping is detected, the remaining data becomes their shared secret key.

Neural network models are defined by weights and biases — the parameters that encode the knowledge learned during training.

- Example: a simple fully connected layer has weights W and biases b , which together define the function $y = f(Wx + b)$.
- In distributed/federated AI scenarios, you may want to **share** these weights between nodes or from a server to edge devices.
- These weights are **sensitive** because:
 1. They **represent proprietary(something private) intellectual property** (trained model).
 2. They can **leak private training data** if intercepted (model inversion attacks).

Thus, transmitting them **securely** is crucial.

What is the goal?

Following the idea we are preventing model theft or tampering during transmission between cloud and edge devices.

Basic Concepts

- Use QKD (e.g., BB84 protocol) to exchange secret keys between sender and receiver.
- Encrypt model weights using the shared quantum key (e.g., via AES or OTP).
- Simulate an eavesdropper(unauthorized party who intercepts) and show that any interception of quantum bits reveals itself via a measurable error rate.

What will be used?

- Python
- Qiskit (For quantum simulation)
- Pytorch (For ML model weights)